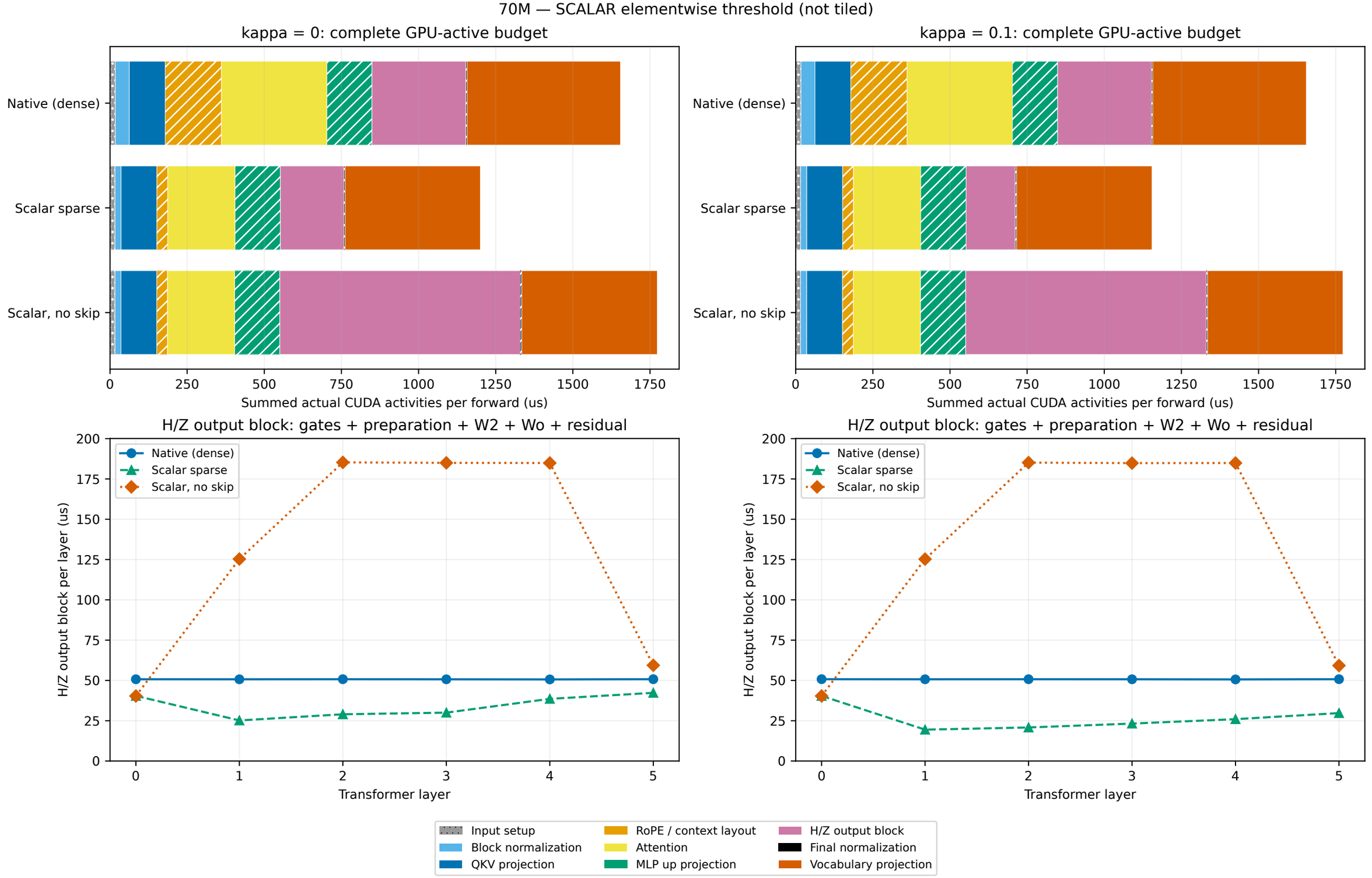

70M — SCALAR elementwise threshold (not tiled)
kappa = 0: complete GPU-active budget
kappa = 0.1: complete GPU-active budget
Native (dense)
Scalar sparse
Scalar, no skip
0
250
500
750
1000
1250
1500
1750
Summed actual CUDA activities per forward (us)
H/Z output block: gates + preparation + W2 + Wo + residual
H/Z output block per layer (us)
Transformer layer
Input setup
Block normalization
QKV projection
RoPE / context layout
Attention
MLP up projection
H/Z output block
Final normalization
Vocabulary projection